Project proposal

Group Name
Date

Throughout this course you will create a data management project that focuses on representing data of your choice in a database. You have been assigned into a group beforehand, please familiarise yourself with your group members and develop a project proposal. The project proposal is submitted for week 4 and should demonstrate what you aim to do in the remaining 8 weeks of the course. You will build a prototype until week 8 and present your final data management product in week 12. For your proposal, aim for a feasible goal and ensure that all team members have a task.

Week	Milestone	Deliverable
0	Group formed	
4	Project proposal	Proposal document (5 pages)
8	Project prototype	Prototype presentation
12	Project presentation	Final presentation, code & report.

Your project proposal should be no longer than 5 pages (plus a title page with group name and members and excluding references) and consists of an introduction (1 page), an overview of the data management problem (1 page) and sketch of your solution (1-2). Lastly, it should contain a timeline (a GANTT chart) that shows the tasks and how each group member will contribute to the different milestones (1-2 pages). Tasks may be described with a sentence each and may be changed later as the project progresses.

1. Choose a topic

There is a wide choice of topics. Three typical ways to find an appropriate topic are:

1. One of you has data or an idea that requires data management
2. There is a current topic in society that could benefit from data management
3. You choose a topic for data management that is suggested by the course

All approaches are valid but please ensure that your choice complies with the requirements listed below. Your data management problem …

- … is relevant to society. Please make your argument in the proposal.
- … is complex (enough). Please explain why you need a database to manage the data and how many relations you roughly expect (UML diagram).
- … can be demonstrated. Either by filling it with a dataset, scrapping some data or creating sufficiently complex synthetic data.
- … has impact. How would your solution improve the situation for society?

If you have no topic of your own, Table 1 lists relevant sectors which require data management to solve issues. You may take up any of the proposed issues or explore these sectors in more depth to find an interesting issue for data management.

TABLE 1: SECTORS IN NEED OF DATA MANAGEMENT.

Sector	Issues
Healthcare	Effective data management is essential for managing vast amounts of patient data, medical records, and billing information to improve patient care, research new treatments, and ensure compliance with privacy regulations.
Finance and Banking	This sector deals with highly sensitive financial and customer data and must adhere to strict regulations. Data management helps in fraud detection, risk management, providing personalized services, and maintaining customer trust.
Retail and E-commerce	Companies use data management to track sales, manage inventory, analyse customer behaviour (e.g., loyalty programs and online activity), predict demand, and enhance the overall customer experience.
Information Technology (IT) and Technology Services	Data management is fundamental to the IT industry itself, supporting database administration, network security, IT asset management, software development, and cloud services.
Government and Public Sector	Agencies handle public records and critical infrastructure data. Effective data management ensures services run smoothly, aids in resource allocation, emergency response, and policymaking, all while safeguarding national security and complying with public information laws.
Logistics and Manufacturing	Data helps these industries optimize operations, manage supply chains, track inventory flow, and improve efficiency in production processes.
Education	Educational institutions use data management for student information systems, academic research, tracking performance metrics, and managing administrative operations.
Media and Entertainment	Data management helps media companies understand viewer preferences, manage large volumes of digital assets (like video files), and provide personalized content recommendations.
Energy	Data management helps manage the complex data involved in energy production, distribution, and consumption.

2. Explain background in the introduction

Your project proposal should highlight how data management is relevant for your chosen topic. Please develop a short introduction in the proposal that explains the issue and how data management can improve the situation. Here is a potential outline what your introduction should cover. It is best to think of the questions that you want to answer for the reader. The questions below are a great guideline for any introduction you will have to write in the future. Therefore, these questions are very general, and you may have to adapt them for your case. In parenthesis, you will find suggestions for adaptation.

1st paragraph: big debate on the topic *(ear opener, where do you encounter this?)*

2nd paragraph: why is the topic important*? (what is the impact of the topic)*

3rd paragraph: What does the (academic) literature say about this? *(what have people tried before?)*

4th paragraph: What is the academic literature on this topic missing? *(what can you do that is new?)*

5th paragraph: What do you do (outline the whole project)? *(what will you do?)*

6th paragraph: What did you find? *(Optional but you can speculate what will be outcome)*

7th paragraph: What are the implications of what you find? *(speculative: what will you be able to do with your data management tool that wasn't possible before?)*

Remember that you are not writing a book or a paper but a proposal. Keep is short and succinct and try to use at least 5 and no more than 10 academic references.

3. Explain your data management problem

In the introduction, you have explained the relevancy of your project and outlined the problem roughly. In this section, you need to get technical and describe why this needs data management and not just research. Typical reasons include scalability (amount of data), complexity (dependencies in data), heterogeneity (different sources and types and formats), but also purpose (create structure to solve a specific problem). Also think of who would use your solution and how your project could improve their situation.

4. Explain your planned solution

Outline the entities and relations you expect in your data management solution. As a default, we will work in PostgreSQL, but you may have reasons to use a different database (e.g. for texts) or an extension (e.g. PostGIS for geographic data or TimescaleDB for temporal data). Explain your choices and justify them were necessary. This plan need not be final but should be the best to your knowledge as of the submission time for the proposal.

5. Propose a timeline and tasks

Your team will consist of 3 to 4 members. Please be mindful that the course covers 10-12 hours of work per week. After lectures, you have 8-10 hours. For self-study and course preparation, you should account for 2 hours per week. That leaves you with 6-8 hours per week to work on the project. At 4 members, you have 24-32 hours per week or a total of 288-384 hours for the project. At 3 members you have 18-24 hours per week or a total of 216-288 hours for the project.

Please propose a timeline that accounts for your actual availability and working hours for the project. Split the tasks fairly across team members and try to ensure that everybody has the same workload in terms of hours. A GANTT chart allows you to visualise the task duration (in hours over the week) which you could colour code by team member to show who is doing what. You may provide short descriptions of each task and the hours needed for the task. Two generated examples are shown in Figure 1 and Figure 2. We will use the GANTT chart to assess both the feasibility and fairness of your team assignments.

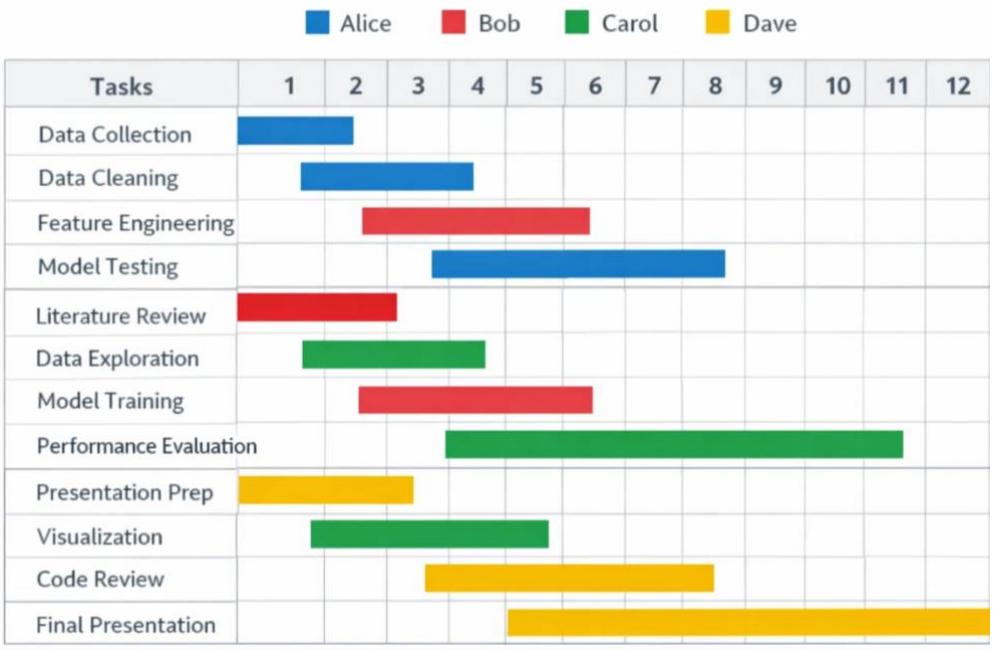

FIGURE 2: IMAGINARY GANTT CHART FOR A PROJECT. COLOURS INDICATE THE TEAM MEMBER, BARS ARE NOT PROPORTIANTE TO HOURS WORKED (YOU COULD DO BETTER THAN THIS EXAMPLE). GENERATED WITH SORA (OPENAI).

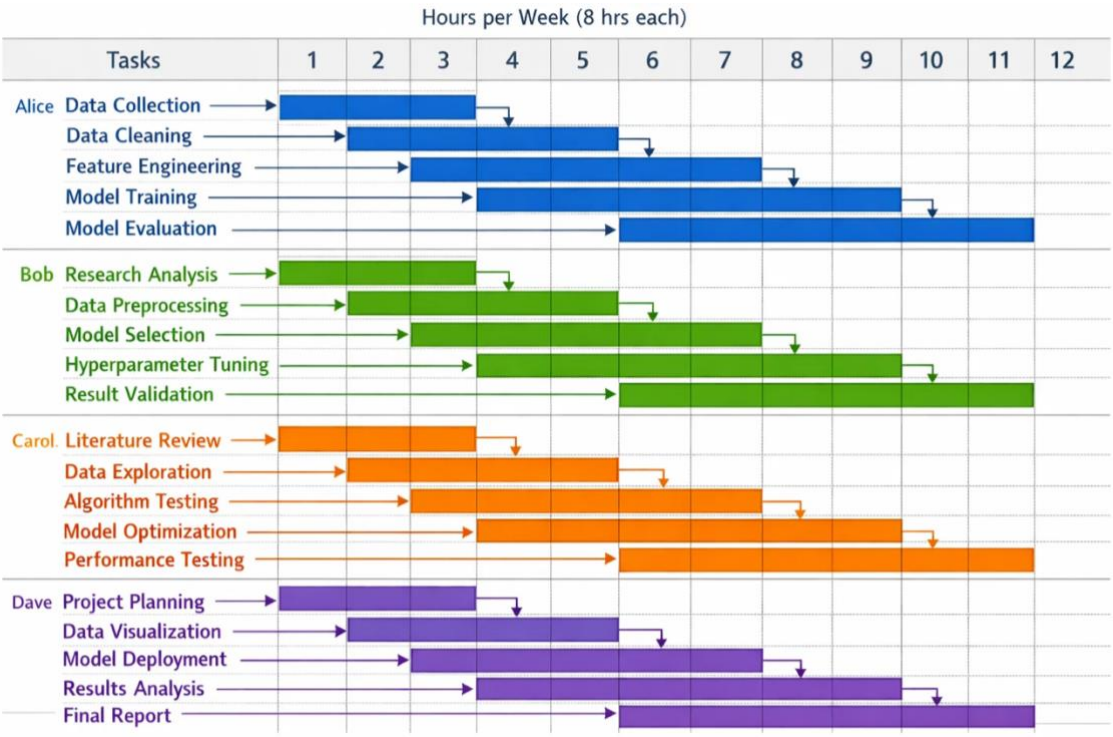

FIGURE 2: IMAGINARY GANTT CHART FOR A PROJECT. COLOURS INDICATE THE TEAM MEMBER, DEPENDENCIES ARE A BIT NON-SENSICAL (YOU CAN PLACE BETTER ARROWS). GENERATED WITH SORA (OPENAI).